Decision-dependent distributionally robust optimization for resilient and sustainable multimodal transportation under endogenous uncertainty

1 Scenario description

Table 1: Characteristics of the disruption scenarios

No.	Scenario description	Probability	Cause of disruption
1	Baseline scenario; no disruption, business-as-usual	0.700	-
2	Flood in Southampton; all rail links to/from Southampton are non-functional, terminal capacity reduced by 0.5, ripple effects on roads	0.015	Flood
3	Flood in Edinburgh and Newcastle; rail links between the two cities are non-functional, geographic dependency leads to road capacity reduction by 0.2	0.010	Flood
4	Flood in Liverpool, Birmingham, and Manchester; rail links between the three cities are non-functional, container operation ripple effects	0.010	Flood
5	Flood in Bristol; all rail links to/from Bristol are non-functional, barge mode affected by geographic influence	0.010	Flood
6	Flood in Milford; all rail links to/from Milford are non-functional	0.010	Flood
7	Flood in Felixstowe, London, and Folkestone; rail links between the three cities are non-functional, major terminal ripple effects	0.015	Flood
8	Flood in Southampton and Bristol; rail links between the two cities are non-functional, interdependencies amplify impacts	0.015	Flood
9	Flood in Southampton; rail links to/from Southampton reduced in capacity by 0.5, terminal operations ripple effects on multi-modes	0.010	Flood
10	Flood in Edinburgh and Newcastle; rail links between the two cities reduced in capacity by 0.5, roads affected by dependencies	0.010	Flood
11	Flood in Liverpool, Birmingham, and Manchester; rail links between the three cities reduced in capacity by 0.5, ripple effects	0.010	Flood
12	Flood in Bristol; rail links to/from Bristol reduced in capacity by 0.5, barge capacity linked reduction	0.010	Flood
13	Flood in Milford; rail links to/from Milford reduced in capacity by 0.5	0.010	Flood
14	Flood in Felixstowe, London, and Folkestone; rail links between the three cities reduced in capacity by 0.5, terminal ripple effects	0.010	Flood
15	London attacked; road links to/from London reduced in capacity by 0.3, rail non-functional, terminal operation ripple effects	0.005	Terrorist attack
16	Birmingham attacked; road links to/from Birmingham reduced in capacity by 0.5, geographic dependencies affect nearby rail	0.005	Terrorist attack
17	Manchester attacked; road links to/from Manchester reduced in capacity by 0.5, container terminal ripple effects	0.005	Terrorist attack
18	Felixstowe terminal equipment failure; capacity reduced by 0.5, ripple effects on rail and truck backlogs	0.005	Equipment failure
19	Southampton port strike; operations halted, multi-modal network ripple effects	0.005	Strike
20	London terminal cyber attack; data disruption, capacity 0.3, supply chain ripple effects	0.002	Cyber attack
21	London-Bristol link bombed; rail capacity reduced by 0.7, geographic dependencies affect roads	0.002	Bombing
22	London-Felixstowe link bombed; rail non-functional, terminal ripple effects	0.002	Bombing
23	London-Folkestone link bombed; rail non-functional	0.002	Bombing
24	London-Southampton link bombed; rail non-functional, ripple effects	0.002	Bombing
25	London-Birmingham link bombed; road capacity reduced by 0.3, interdependencies amplify	0.002	Bombing